

TRYASIL VEGSTASJON, Norway

WMO: 013970

Lat: 61.2942N

Lon: 12.2717E

Elev: 361

StdP: 97.06

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.1	-22.6	-27.9	0.3	-25.0	-25.2	0.4	-22.6	6.4	-0.2	5.3	-0.5	0.6	130	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	26.5	15.3	24.2	15.0	22.0	14.1	17.2	22.8	16.1	21.2	15.1	20.0	2.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.2	11.3	18.4	14.1	10.5	17.4	13.1	9.8	16.5	49.6	22.6	46.2	21.1	43.3	19.9	20.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.3	5.3	4.4	-28.3	29.0	2.0	2.5	-29.8	30.7	-31.0	32.2	-32.1	33.6	-33.6	35.3
			-28.5	18.5	2.0	1.4	-29.9	19.5	-31.1	20.4	-32.2	21.1	-33.6	22.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-9.0	-7.4	-2.9	2.7	7.7	12.2	15.1	13.0	9.0	2.7	-2.5	-7.4	
	DBStd	9.41	6.59	5.89	5.01	3.25	3.95	3.27	3.05	2.70	3.07	3.80	4.80	6.71	
	HDD10.0	3007	589	487	401	220	94	13	1	6	54	227	376	539	
	HDD18.3	5674	847	720	659	470	329	185	109	166	281	483	626	798	
	CDD10.0	384	0	0	0	1	23	80	157	99	22	2	0	0	
	CDD18.3	9	0	0	0	0	0	1	8	1	0	0	0	0	
	CDH23.3	274	0	0	0	0	17	81	158	18	0	0	0	0	
	CDH26.7	44	0	0	0	0	0	13	31	1	0	0	0	0	
Wind	WSAvg	1.9	1.5	1.6	1.9	2.1	2.2	2.4	2.0	1.8	1.9	1.7	1.6	1.5	
Precipitation	PrecAvg	829	56	41	34	45	75	84	93	109	79	84	74	61	
	PrecMax	1124	127	98	71	107	162	145	175	151	145	186	214	137	
	PrecMin	683	9	10	4	10	24	47	25	44	27	16	28	12	
	PrecStd	106	28	18	17	25	32	27	40	28	35	43	40	29	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	5.8	12.1	18.0	25.7	28.4	29.5	25.6	20.8	15.0	9.1	5.9	
		MCWB	2.5	2.8	6.2	9.7	13.8	14.3	17.4	15.8	13.9	11.5	7.9	4.3	
	2%	DB	2.3	3.0	8.3	15.5	22.4	25.9	27.2	22.8	18.5	11.9	7.0	4.1	
		MCWB	1.1	1.5	3.9	8.0	12.6	13.9	16.6	14.9	12.6	9.2	6.0	2.9	
	5%	DB	0.8	1.5	5.9	12.6	19.8	22.9	25.0	20.9	16.6	10.3	5.2	2.5	
		MCWB	-0.1	0.4	2.2	6.4	11.5	13.7	15.7	14.2	11.9	8.1	4.5	1.5	
	10%	DB	0.0	0.4	3.8	9.7	16.3	20.2	22.8	19.0	14.7	8.7	3.4	1.0	
		MCWB	-0.6	-0.3	1.2	4.7	9.8	12.3	15.0	13.5	11.1	7.1	2.6	0.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	3.4	6.4	10.2	15.4	17.1	19.1	17.6	15.1	12.3	8.4	4.7
			MCDB	3.5	4.9	11.3	16.8	21.5	22.3	24.7	21.3	18.3	14.0	9.1	5.6
2%		WB	1.4	1.7	4.1	8.4	13.8	15.3	17.8	16.4	13.7	10.0	6.2	3.2	
		MCDB	2.1	2.8	7.4	14.3	20.9	21.7	24.8	20.0	16.6	11.6	7.0	4.0	
5%		WB	0.2	0.6	2.7	6.7	12.4	14.3	16.8	15.3	12.7	8.5	4.7	1.6	
		MCDB	0.7	1.3	5.1	12.2	18.1	21.2	22.9	19.1	15.6	9.8	5.2	2.3	
10%		WB	-0.7	-0.2	1.5	5.1	10.6	13.2	15.7	14.3	11.6	7.2	2.8	0.4	
		MCDB	-0.1	0.4	3.6	9.2	15.6	19.1	21.3	17.8	14.2	8.6	3.3	0.9	
Mean Daily Temperature Range		5% DB	MDBR	7.3	8.9	11.2	11.6	11.9	12.2	11.7	10.8	10.5	7.8	5.6	7.0
			MCDBR	6.3	8.8	13.2	17.3	18.9	18.2	16.5	14.5	13.9	10.3	6.7	6.9
	MCWBR		5.7	7.2	9.0	10.5	9.8	8.5	7.4	7.8	8.6	7.6	5.9	6.3	
	5% WB	MCDBR	5.7	7.6	11.8	16.5	15.4	14.8	14.0	11.4	11.8	8.4	6.1	6.7	
		MCWBR	5.3	6.3	8.2	10.2	8.5	7.4	6.7	6.6	7.9	6.7	5.5	6.2	
Clear-Sky Solar Irradiance	taub	0.222	0.271	0.301	0.339	0.346	0.339	0.358	0.348	0.317	0.301	0.241	N/A		
	taud	2.134	2.249	2.253	2.303	2.434	2.482	2.441	2.460	2.534	2.479	2.225	N/A		
	Ebn at Noon	525	718	818	853	878	890	860	840	810	690	494	N/A		
	Edh at Noon	36	63	90	105	100	97	99	89	69	52	33	N/A		
All-Sky Solar Radiation	RadAvg	0.28	0.95	2.50	4.09	4.70	5.24	4.74	3.69	2.41	1.08	0.34	0.16		
	RadStd	0.04	0.18	0.25	0.42	0.45	0.44	0.63	0.34	0.29	0.19	0.09	0.02		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page